ORE AMENDMENT APPENDIX

APPENDIX: COMPLETE USER STUDY DATA COLLECTION INSTRUMENTS

Study Title:	Evaluating Explainable AI (XAI) Dashboards for Insider Threat Detection
Protocol Layout:	Counterbalanced Within-Subject Evaluation Design (N = 36)
Data Platforms:	Local Embedded Testing Software (Dashboard) & Secure Institutional Qualtrics Environment
Date of Revision:	July 2026 (Updated for ORE Voluntary Participation Standards)

PART 1: LOCAL EMBEDDED DASHBOARD INSTRUMENTS

Administration Protocol: These data capture fields are embedded directly within the dashboard's user triage panel. Quantitative decision logs, psychometric workload scales, and open-ended text fields are presented following each of the 4 scenario trials. Complete inputs or explicit bypass entries ('Pass' or 'N/A') are programmatically required prior to allowing file compilation and forensic export to ensure local database stability without forcing substantive text submission.

I. Operational Decision Quality Logs (Categorical Selectors)

1. Individual Daily Alarm Verdicts (Prompted 5 times per scenario)

- **Prompt:** Select Operational Disposition Verdict:
- **Response Options (Binary Button Choice):**
  - Confirm Threat (Malicious Threat)
  - Dismiss Alarm (Benign Deviation)

2. Case Closure Classification (Prompted 1 time per scenario)

- **Label Prompt:** Case Closure Classification:
- **Response Options (Dropdown Select Matrix):**

Classification Option	S1	S2	S3	S4	S5
Removable Storage Exfiltration / IP Loss					
Active Competitor Solicitation / Corporate Fraud					
Host Malicious Misuse via Keylogging Agent					
Unauthorized Target Workspace Hijacking					

Classification Option	S1	S2	S3	S4	S5
Multi-Vector Overlapping Profile (INVALID SELECTION - DESIGN ARTIFACT)					

3. Decision Certainty Index (Prompted 1 time per scenario)

- **Label Prompt:** Analyst Confidence Level:
- **Response Options (Dropdown Select Matrix):**

Confidence Level	Score
Highly Confident (Certainty)	5
Moderately Confident (High Probability)	4
Standard Confidence (Indeterminate)	3
Low Confidence (Speculative)	2
Skeptical / Highly Uncertain	1

4. Case Justification Free-Text Field (Prompted 1 time per scenario)

- **Label Prompt:** Analyst Case Justification Narrative (3-line entry profile):
- **Response Constraint & Placeholder:** Open text box entry field.

> *"Enter tactical diagnostic summary, relevant behavioral updates, and mitigation sign-off rationale details... (If you do not wish to answer, you may simply enter 'Pass' or 'N/A' to bypass this question and proceed.)"*

Voluntary Field Bypass Enabled

II. Subjective Task Load Metrics (NASA-TLX Short-Form Sliders)

> **Scale Architecture:** 21-point discrete integer slider scale (1 = Low, 11 = Moderate, 21 = High). Interactive tooltips display automatically upon mouse hover to clarify metrics uniformly.

1. Mental Demand Dimension

- **Label Prompt:** Mental Demand (How much mental/perceptual activity was required?)
- **Hover Tooltip Content:**

> **NASA-TLX: Mental Demand**
> **Definition:** How much mental and perceptual activity was required to complete this triage sequence (e.g., thinking, deciding, calculating, or searching)?
> **Scale Bounds:** 1 = Straightforward/Simple Context vs. 21 = Extreme Cognitive Complexity/Filtering.

2. Frustration Level Dimension

- **Label Prompt:** Frustration Level (How insecure, discouraged, or annoyed did you feel?)
- **Hover Tooltip Content:**

> **NASA-TLX: Frustration Level**
> **Definition:** How insecure, discouraged, irritated, stressed, or annoyed did you feel during this specific evaluation trial?
> **Scale Bounds:** 1 = Smooth and highly satisfying triage vs. 21 = Technical friction and intense analytical stress.

III. Immediate Localized Strategy Tracking (Dynamic Free-Text Fields)

> **Conditional Software Logic:** The dashboard presents only one of the following prompts matching the active interface architecture layout under evaluation.

Frame A Question: Design A Condition (Traditional Baseline Workflow)

- **Prompt Text:** Strategy Feedback: To what extent did the calculated Z-scores in the Feature Table cross-validate or replace your manual searching of the Raw Action Logs?
- **Response Constraint:** Text box container (minimum 1 sentence required; participants may enter 'Pass' or 'N/A' to bypass without penalty). **Voluntary Bypass Active**

Frame B Question: Design B Condition (Explainable AI Workflow)

- **Prompt Text:** Strategy Feedback: Did you formulate your thesis directly from the red/blue SHAP waterfall attributions, or did you still feel compelled to open raw logs to verify your decision? Why?
- **Response Constraint:** Text box container (minimum 1 sentence required; participants may enter 'Pass' or 'N/A' to bypass without penalty). **Voluntary Bypass Active**

PART 2: GLOBAL RETROSPECTIVE QUALTRICS INSTRUMENTS

Administration Protocol: This independent questionnaire is executed once at the conclusion of Phase 3 (Post-Study Quantitative Metrics) after the participant has experienced both layouts completely. In compliance with institutional ethics requirements for voluntary participation, all hard "Force Response" configurations are deactivated and replaced with standard "Request Response" reminders, ensuring participants can freely skip any prompt.

I. Demographics and Balanced Cohort Filtering

1. Anonymous Participant Mapping

- **Prompt:** Enter your assigned SOC Analyst ID (1-36):
- **Response Constraint:** Character field with numeric integer enforcement bounds. **Voluntary Skip Allowed**

2. Counterbalancing Order Validation

- **Prompt:** Select your assigned counterbalancing group:
- **Response Options:**

  - **Group 1:** Evaluated Scenarios 1 & 3 on Design A Baseline | Scenarios 2 & 4 on Design B SHAP Visuals
  - **Group 2:** Evaluated Scenarios 1 & 3 on Design B SHAP Visuals | Scenarios 2 & 4 on Design A Baseline

  Voluntary Skip Allowed

3. Skill Stratification Control

- **Prompt:** Select your expertise cohort profile:
- **Response Options:**

  - **Junior Cohort:** Software/Systems/Computer engineering student without advanced security courses or co-op experience
  - **Senior Cohort:** Graduate safety/security student, upper-year undergraduate with CS/ECE 458, or min. 1 cybersecurity co-op term

  Voluntary Skip Allowed

II. Retrospective Comparative System Usability Scale (SUS Matrix Table)

Layout Architecture: Side-by-side comparative table layout prompting the participant to grade both dashboard interface experiences concurrently to mitigate contrast bias. Choices track a 5-point Likert Scale: 1 (Strongly Disagree), 2, 3, 4, 5 (Strongly Agree). **Standard 'Request Response' Reminder Only — Skipping Permitted**

System Usability Scale (SUS) Core Items	Design A (Baseline)					Design B (SHAP Visuals)				
	1	2	3	4	5	1	2	3	4	5
1. I think that I would like to use this system frequently.										
2. I found the system unnecessarily complex.										
3. I thought the system was easy to use.										
4. I think that I would need the support of a technical person to be able to use this system.										
5. I found the various functions in this system were well integrated.										
6. I thought there was too much inconsistency in this system.										
7. I would imagine that most people would learn to use this system very quickly.										
8. I found the system very cumbersome to use.										
9. I felt very confident using the system.										
10. I needed to learn a lot of things before I could get going with this system.										

III. Global Qualitative Retrospective Questions (Essay Responses)

All essay fields utilize standard "Request Response" reminders to preserve complete participant autonomy. Content skip is fully permitted.

1. **Question 1 (Workflow Delta Strategy Assessment)**
   - **Prompt Text:** You evaluated identical attack signatures across different timelines using both traditional Z-score tables (Design A) and SHAP visual attribution plots (Design B). How did your workflow or strategy change when transitioning between these two systems? For instance, did you find yourself looking for different types of evidence?
   - **Response Constraint:** Essay text window framework. **Voluntary Skip Allowed**

2. **Question 2 (Generative Explanation Interactive Utility)**
   - **Prompt Text:** When using the interactive Gemini LLM chatbot panel in Design B, did the AI-generated text explanations accurately clarify the local waterfall or global beeswarm distribution profiles? Did interacting with the chatbot alter your confidence level when committing to a final case closure disposition?
   - **Response Constraint:** Essay text window framework. **Voluntary Skip Allowed**

3. **Question 3 (Cognitive Gap Bridging and Skill Stratification)**
   - **Prompt Text:** As a proxy analyst, did you feel that the SHAP overlays (waterfall and beeswarm graphs) helped bridge any gaps in your specialized cybersecurity domain knowledge, or did they introduce an additional layer of visual complexity that required extra effort to learn?
   - **Response Constraint:** Essay text window framework. **Voluntary Skip Allowed**